

Perdiction Results

2024-10-02

Data Summary:

- **Number of observations for bill passage predictions:** 40,000.
- **Number of observations for bill summary completion:** 39,993. A total of 37 responses exceeded the token limit of 750, and 7 null responses were dropped.

Results

Bill Passage

Table 1: True Bill Passage Rates

Chamber	True Passage Rate
Pass House	0.074
Pass Senate	0.058

Table 2: Performance Evaluation of LLM Predictions of Bill Passage

Predicted Outcome	Model	Prompt	Accuracy	TPR	FPR
Pass House	GPT-3.5	w/o date restriction	0.180	0.976	0.884
Pass House	GPT-3.5	w/ date restriction	0.159	0.987	0.908
Pass House	GPT-4o	w/o date restriction	0.446	0.866	0.588
Pass House	GPT-4o	w/ date restriction	0.241	0.981	0.818
Pass Senate	GPT-3.5	w/o date restriction	0.275	0.963	0.767
Pass Senate	GPT-3.5	w/ date restriction	0.252	0.972	0.792
Pass Senate	GPT-4o	w/o date restriction	0.486	0.885	0.539
Pass Senate	GPT-4o	w/ date restriction	0.300	0.969	0.741

Bill Summary Completion

Table 3: Performance Evaluation of LLM Completion of Bill Summaries (Non-Embedded)

Model	Prompt	Average
GPT-3.5	w/o date restriction	0.014
GPT-3.5	w/ date restriction	0.015
GPT-4o	w/o date restriction	0.042
GPT-4o	w/ date restriction	0.038

Table 4: Performance Evaluation of LLM Completion of Bill Summaries (Embedded)

Metric	Model	Prompt	Average	Baseline	Relative to Baseline
Cosine Similarity	GPT-3.5	w/o date restriction	0.489	0.244	1.000
Cosine Similarity	GPT-3.5	w/ date restriction	0.500	0.243	1.053
Cosine Similarity	GPT-4o	w/o date restriction	0.537	0.244	1.200
Cosine Similarity	GPT-4o	w/ date restriction	0.542	0.244	1.224
Euclidean Distance	GPT-3.5	w/o date restriction	0.987	1.227	0.196
Euclidean Distance	GPT-3.5	w/ date restriction	0.975	1.228	0.206
Euclidean Distance	GPT-4o	w/o date restriction	0.920	1.228	0.250
Euclidean Distance	GPT-4o	w/ date restriction	0.917	1.228	0.254